REPORT

Course Code: CSE 4408

Course Name: System Analysis and Design

Lab Number: 5

Project Title: BashayJabo

Group: Les Misérables

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1.1 Stakeholder Interviews

Stakeholders & Interview Objectives

For our interview process, we have decided to split our general target stakeholders (students of IUT) into 3 distinct categories depending on their frequency and role in commuting. Depending on this categorisation, we will be able to extract role-specific information that is crucial to our requirement elicitation.

Stakeholder	Role Description	Interview Objective
Frequently Commuting Students	Students, usually non-residential, who commute to and from campus daily.	Understand daily needs, route patterns, and transportation preferences.
Occasional Travelers	Permanent residential students, who only travel home for emergencies or during long breaks.	Explore flexibility such as public transport coordination or even flight ticketing, and other one-time needs.
Regular Organisers	Students who have experience with and often travel in groups.	Identify pain points in coordinating shared rides, especially via informal messaging.

Interview Guide

a) **Frequently Commuting Students**
   - **Intro Script:** “Thank you for joining. We're a team designing a student-friendly ride-sharing system just for IUT. Since you commute to and from campus on a daily basis, your input would be invaluable to us in making your experience more convenient.”
   - **Structure and Outline of Questioning:** Funnel - start from broad questions that aim to explore their general experiences, habits, and pain points before narrowing the focus down to introducing the idea of ride-sharing.
   - **Questions:**
      i. Can you describe your daily commute to IUT? What challenges have you faced? (Open)
      ii. What factors influence your choice to share or travel solo? (Probing)
      iii. Would you be comfortable sharing rides with fellow students through a mobile app? (Closed)
   - **Closing Script:** "Thank you for your time! We really appreciate your feedback. Your feedback will aid us in developing a better commuting experience for IUT students. We may reach out again later to test a prototype, would that be okay with you?"

b) **Occasional Travelers**
   - **Intro Script:** “Thank you for joining. We're a team designing a student-friendly ride-sharing system just for IUT. Since you don’t commute on a regular basis, we want to understand your specific needs when you do so, especially when travelling out of Dhaka.”
   - **Structure and Outline of Questioning:** Funnel - starting from broad questions that understand the context of and frustrations in their occasional commutes, before narrowing the focus down to introducing the idea of ride-sharing.

- **Questions:**
    1. Can you describe your journey to Dhaka/IUT? What challenges have you faced? (Open)
    2. Do you find it more convenient to travel alone or in coordinated groups? (Probing)
    3. Would you be more comfortable on long journeys if you could choose your preferred travel-mates? (Closed)
- **Closing Script:** "Thank you for your time! We really appreciate your feedback. This was crucial for us in providing a more comfortable travel experience for IUT students. We may reach out again later to test a prototype, would that be okay with you?"

c) **Regular Organisers**

- **Intro Script:** “Thank you for joining. We're a team designing a student-friendly ride-sharing system just for IUT. Since you are already familiar with coordinated ride-sharing, we want to know how we can make the process easier for you.”
- **Structure and Outline of Questioning:** Diamond - start with closed questions understanding their methods of coordination, before opening up to their broader points of struggle and past experiences. End with closed questions confirming our findings.
- **Questions:**
    1. What is your usual mode of transport? Do you currently use any ride-booking apps? (Closed)
    2. What challenges do you often face in coordinating your travel groups? (Open)
    3. So would you prefer a system that provides location-based suggestions according to your preferences? (Closed)
- **Closing Script:** "Thank you for your time! We really appreciate your feedback. Your input is extremely helpful for us in designing a better commute coordination experience for IUT students. We may reach out again later to test a prototype, would that be okay with you?"

Interview Report – Frequently Commuting Student (Simulated)

Context

- Interviewee Profile: 2nd-year female student from CSE Department, lives in Uttara.
- Commuting Pattern: Travels to IUT 5 days a week.
- Method: In-person, semi-structured interview (~10 minutes).
- Date: [21/06/2025]

Interview Objectives and Key Findings

1. Understand current commuting methods and challenges

- Travels by Uber or shared CNG depending on the day.
- Often coordinates rides informally through Messenger groups.
- Finds it stressful to confirm who's coming and when.

Notable Quote: "It's always last-minute. Someone cancels, and we end up paying more or getting late."

2. Identify user expectations for a ride-sharing system

- Wants to see only available rides that match class timing.
- Prefers known people or students from the same department.
- Would like auto-split fare option and shared ride visibility.

Notable Quote: "If I can ride with someone from CSE and we know each other, I'd feel safer, and my journey would be smoother."

3. Discover issues with existing apps

- Uses Uber often but dislikes the randomness of who shares.
- Finds ride-hailing apps expensive when traveling alone.
- Wishes there was a group ride option built for students.

Notable Quote: "I wish Uber let me invite specific people to ride with me."

4. Gauge comfort with app features and integrations

- Comfortable using a mobile app; prefers it over web.
- Supports integration with Uber or Shohoz if student filters are applied.
- Willing to give feedback or ratings after rides.

Conclusion: The interviewee strongly supports the idea of a structured, student-only ride-sharing platform with personalized filtering and cost-saving features. The insights align with common student concerns like unpredictability, safety, and peer compatibility.

1.2 User Questionnaire

Purpose, Confidentiality, and Instructions

The purpose of this questionnaire was to collect feedback from IUT students on their commuting habits, ride-sharing preferences and expectations from a digital coordination system. All responses were collected anonymously and used solely for academic and design analysis. Participants were encouraged to answer honestly and only select options that reflected their actual behavior and preferences.

Questionnaire Structure

- **Brief Introduction:** “We are a team designing a student-friendly ride-sharing system for IUT, and would like to know about your commuting habits to and from campus. All information will be kept confidential - this is simply for us to better understand your needs.”
- **Sample Questions:**
    1. How often do you travel to/from IUT? (MCQ)

       ☐ Daily ☐ Weekly ☐ During holidays ☐ During semester break
    2. What mode of transport do you usually take? (Checkboxes)

       ☐ Ride-sharing apps ☐ Bus ☐ Metro ☐ CNG ☐ Private vehicle
    3. How often do you coordinate rides with others? (Likert scale)

       ☐ Yes, regularly ☐ Sometimes ☐ No, I always travel alone
    4. Which preferences would you like to set in a ride-sharing app?

       ☐ Gender ☐ Department ☐ Batchmates ☐ Transport type
    5. Which ride-sharing apps do you currently use? (Checkboxes)

       ☐ Uber ☐ Pathao ☐ inDrive ☐ Others
    6. What kind of ride booking and sharing system would you prefer?

       ☐ Mobile app ☐ Web app ☐ Existing apps ☐ Manual coordination
    7. How much are you willing to share a ride with others? (Likert scale)

       ☐ Not at all interested

       ☐ Slightly interested

       ☐ Interested

       ☐ Very interested

8. What shortcomings do existing ride booking apps have as per your opinion? (Open)
9. What would make you more likely to use a ride-sharing app for IUT? (Open)
10. Describe a frustrating or inconvenient experience you've had with ride-sharing or commuting. (Open)

Target Respondents, Delivery, and Timing Strategy

- Target Group: IUT students from various departments and years who commute to and from campus.
- Delivery Method: Google Form shared in class group chats, dorm groups, and personal networks.
- Timing Strategy: The questionnaire was initially distributed on a Friday evening, allowing students to respond over the weekend when they are relatively free from academic pressure. A reminder message was sent on Sunday evening to encourage participation before the new academic week began, maximizing the chances of thoughtful responses.
- Incentive: Voluntary, no reward emphasis on helping improve student life.

Participant Summary and Response Rates

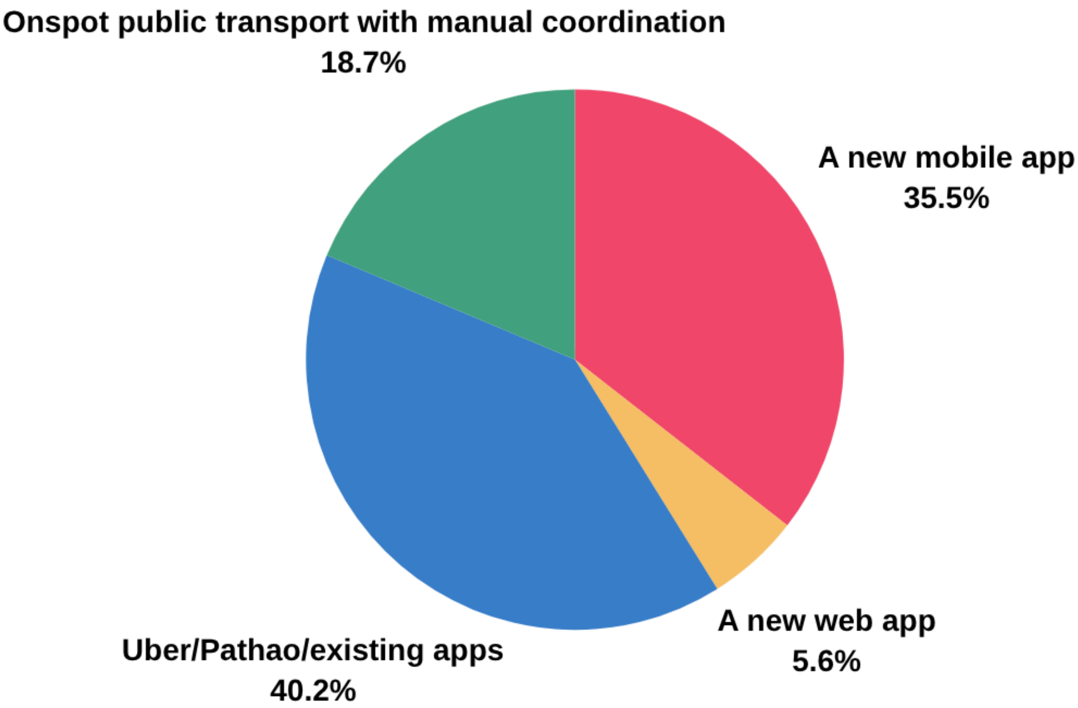
- Total Responses Received: 62
- Response Rate (estimated reach: ~150 students): 41.33%

a) Tabular Summary of Closed-Ended Questions

Question	Most Chosen Option	% of Participants
Travel Frequency	Weekly	52.3%
Transport choice	Bus	41.2%
Willing to Share Rides	Interested	64.67%
Preferred ride sharing filter	Transport type	34.68%
App Type	Mobile app	61%
Preferred ride booking and sharing system	Existing apps	40.2%
Preferred ride-sharing app	Uber	51%

b) **Visual Representation of Response Data**

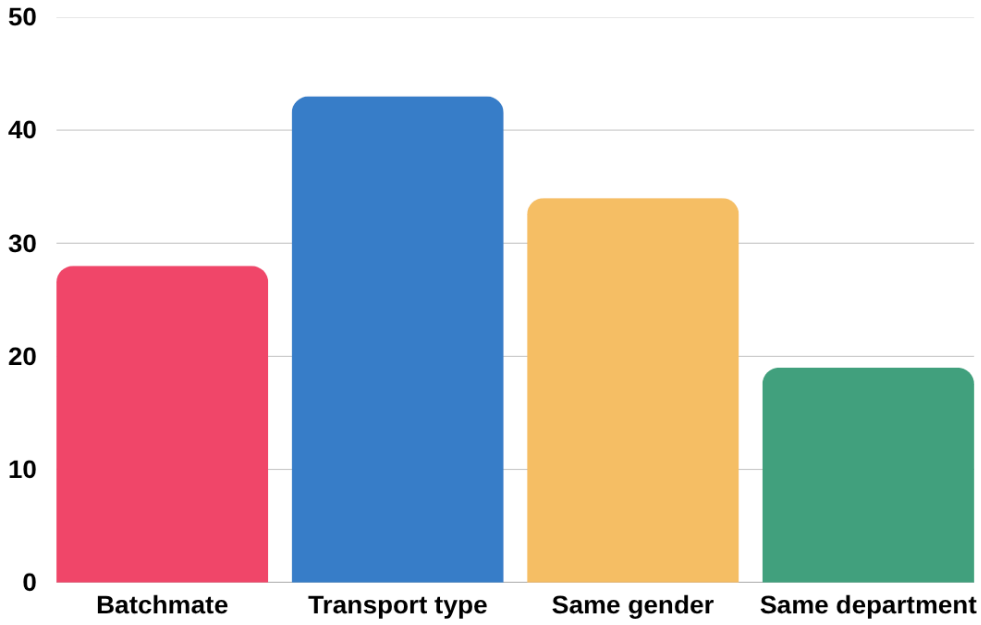
Q. What kind of centralised ride booking and sharing system would you prefer?



Observation:

- While existing apps dominate, more than 41% (mobile + web) want new solutions, showing real interest in better alternatives.
- Mobile-first development is critical; students clearly prefer smartphones over web-based interfaces.
- Manual coordination is still used for informal communication but could be streamlined or phased out with better tech.

Q. If customization was possible for ride sharing, what filters would you use?



Observation:

- Filters based on transport type and same gender are the most wanted.
- Social familiarity (batch, department) is appreciated but slightly less prioritized.
- These results guide feature prioritization when building filter options into the app.

c) Open-Ended Feedback: Thematic Summary

1. **What shortcomings do existing ride booking apps have as per your opinion?**
   - **Common Themes Identified:**
     - Lack of Student-Specific Features: Most existing apps (like Uber or Pathao) don't allow filtering for student-only or known riders, making them feel less safe or personal.
     - High Cost: Several students mentioned that rides are often expensive, especially when traveling alone from remote areas like Uttara or Tongi.
     - Unreliable Availability: Users noted difficulties in finding rides during peak times or near IUT, particularly early morning or late evening.
     - No Group Coordination Option: Apps don't support coordinated group booking, meaning students have to rely on messaging apps separately.
   - **Sample Comments:**
     - "I wish Uber had a group travel feature where I could share rides with friends or known people."
     - "The prices go up at the worst times and it becomes hard to travel alone with such high prices, and I can't always find a ride."

2. **What would make you more likely to use an IUT ride-sharing app?**
   - **Common Themes Identified:**
     - Partner Filters: Students overwhelmingly want the ability to choose or filter ride partners (same gender, department etc).
     - Recurring Ride Setup: A feature to schedule weekly rides (e.g., for the same 8:00 AM class every day) was mentioned frequently.
     - Fare Splitting: Automatic fare splitting was a popular request to avoid awkward calculations or payment follow-ups.
     - Feedback System: A built-in rating or feedback system for ride partners was seen as important for safety and accountability.

- **Sample Comments:**
  - “If I can ride with people who match my preferences like same gender or department and split the fare easily, I’d definitely use it.”
  - “Being able to schedule rides for the whole week at once would save time.”

3. **Describe a frustrating or inconvenient experience you’ve had with ride-sharing or commuting.**
   - **Common Themes Identified:**
     - Communication Issues: It’s hard to coordinate exact times or pickup points through informal chats.
     - Mismatch in Ride Goals: Sometimes, students going in slightly different directions agree to share rides, but the detours make others late.
   - **Sample Comments:**
     - "Sometimes we get confused about the pickup order, which causes delays for everyone."

1.3 Requirement Elicitation

From our various methods of user data collection, we gained a clear perspective of the features and functionality of our system as supported by user feedback data.

a) **Functional Requirements:**
   - The system shall allow users to create new rides.
   - The system shall allow users to join existing ride requests.
   - The system will match users based on location, time, and preferences.
   - The system will connect with Uber/Pathao APIs, since the two ride-booking apps were most common amongst students (51% considered Uber their preferred app).

b) **Non-Functional Requirements:**
   - The system UI design shall be responsive for mobile use, as 61% of users would be willing to use the system if it was a mobile app.
   - The system shall complete ride-matching within 5 seconds.
   - The system shall securely store login and payment data.

c) **User Requirements:**
   - Students want to control who joins their rides (filters: same gender, department, etc). (34.68% prefer rides filtered by transport type, 27.42% prefer rides filtered by gender and 22.58% prefer rides filtered by batch).
   - Easy interface for frequent commuters.
   - Anonymous feedback option for ride experiences.

d) **Business Requirements:**
   - The system should reduce the average transport cost per student.
   - It should support peak-hour load balancing.

e) **Technical Constraints:**
   - Dependent on third-party API availability and limits.
   - Requires university email-based login system.

1.4 Other Interactive Methods

Storytelling / User Scenarios

When designing this system, focusing only on high-level features or technical goals is not enough. Many crucial user expectations, pain points, and edge cases remain hidden unless we dig deeper into the actual experiences of the people who will use the system. Unlike abstract feature requests, real stories are rooted in lived experience. They bring out edge cases (what happens when the system fails or behaves unexpectedly), emotional responses (like trust, frustration, or relief), and workarounds users create when the system doesn't meet their needs. These elements will help us build solutions that are not just technically functional but realistically usable, empathetic, and robust.

Context: While conducting interviews, we encouraged participants to share personal commuting stories. One student described waiting over 30 minutes for a shared CNG due to poor coordination. He had no clear idea who else was heading to the same destination, leading to frustration and uncertainty.

Finding: This anecdote revealed several implicit requirements:

- A live ride/partner availability view is crucial.
- Students want real-time ride status updates. If a ride has already left, it should no longer be visible to ride-seekers.

JAD Workshop Feasibility

- **Feasibility:** Assessing the feasibility of conducting a Joint Application Development (JAD) workshop for this project reveals several constraints that suggest alternative methods may be more effective. The system's complexity is moderate - complex enough to require user input but not so intricate that it demands real-time, high-stakes group modeling.

Feasibility Factor	Assessment
Stakeholder Availability	Limited due to tight academic schedules
Collaboration Culture	Informal, but not structured for long, intensive sessions
System Complexity	Moderate, suitable for interviews and surveys for now
Need for Diverse Brainstorming	Present, but managed through informal chats and voice notes
Logistical Constraints	No dedicated admin/sponsor to coordinate a JAD session

- **Alternative chosen:** Given these factors, we decided to rely on more flexible and scalable methods such as structured interviews, online surveys, and small group discussions rather than a full JAD session at this stage of the project.